DINOSAURS

FOR 150 MILLION YEARS, from the Triassic Period until the end of the Cretaceous Period, 65 million years ago, dinosaurs lived on Earth. Their remains have been discovered in every continent including Antarctica. They formed a varied group of land-living reptiles. People who study prehistoric life, called palaeontologists, divide them into two main groups – the Ornithischia and the Saurischia. There were meat-eating and plant-eating dinosaurs. Some dinosaurs, were huge; others were only the size of chickens.

Iguanodon skull

Iguanodon skull
Gideon Mantell, an English doctor, named *Iguanodon* in 1825, noting the similarity between its teeth and those of the modern iguana. *Iguanodon's* teeth were shaped to fit tightly together. They wore down as the dinosaur chewed its food of tough vegetation with the help of a hinged jaw.

Iguanodon tooth

Iguanodon

Iguanodon
This was one of the first dinosaurs to be discovered. Modern reconstructions give it an outstretched tail and forelimbs that can reach the ground.

Iguanodon foot

Iguanodon foot
The feet of *Iguanodon* had small hooves on the toes instead of claws, and would have made recognizable three-toed prints with rounded digits. *Iguanodon* probably walked on its toes, which, therefore, had to be strong to carry the animal's great weight.

Ornithischians

The Ornithischia, or bird-hipped dinosaurs, such as *Iguanodon*, were all herbivorous. They had a huge number of teeth – *Corythosaurus* had 2,000 – and a hinged upper jaw that allowed them to chew.

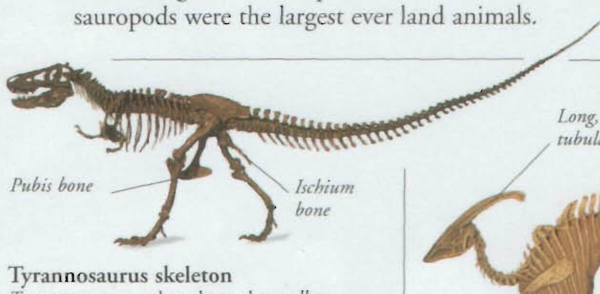
Saurischians

The Saurischia, or lizard-hipped dinosaurs, include the meat-eating theropods, which walked on two legs, such as *Tyrannosaurus*, and the plant-eating sauropods, which walked on four legs, such as *Diplodocus*. The sauropods were the largest ever land animals.

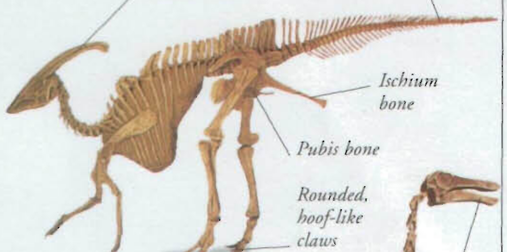
Tyrannosaurus tooth
Carnivorous dinosaurs had curved, pointed teeth. The sharp edges often had serrations, which helped the dinosaurs to slice through skin and meat Palaeontologists still have to be careful when handling these teeth.

Tyrannosaurus

Tyrannosaurus *tore off the flesh of its prey with its teeth and claws.*



Tyrannosaurus skeleton
Tyrannosaurus may have hunted as well as scavenged on other dinosaurs. It had a massive skull with powerful jaws, supported by a short, flexible neck. This flexibility allowed the animal to twist its head around to wrench flesh from its prey.

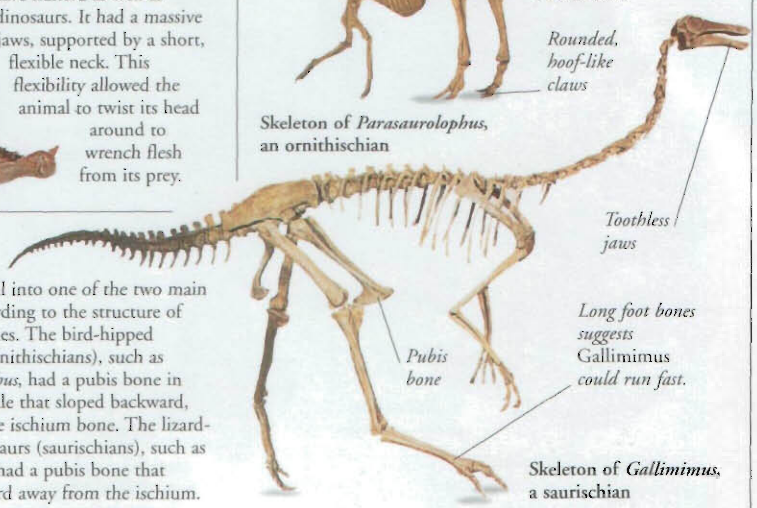


Skeleton of *Parasaurolophus*, an ornithischian

Tyrannosaurus
Although not thought to be the largest of the carnivorous dinosaurs, *Tyrannosaurus* was still an extremely fearsome predator. It walked on its hind legs with its back level and head raised. It could run very fast, its tail balancing the weight of its huge heavy body.

Fossil dung
Preserved pieces of dung are called coprolites. They contain the remains of what dinosaurs ate, such as bone fragments, fish scales, or plant remains. Scientists can study these to find out about the diet of dinosaurs.

Hips
Dinosaurs fall into one of the two main groups, according to the structure of their hip bones. The bird-hipped dinosaurs (ornithischians), such as *Parasaurolophus*, had a pubis bone in their hip girdle that sloped backward, parallel to the ischium bone. The lizard-hipped dinosaurs (saurischians), such as *Gallimimus*, had a pubis bone that sloped forward away from the ischium.



Skeleton of *Gallimimus*, a saurischian